Complex Numbers: De Moivre's Theorem

Questions 1–24

Question 1. Use De Moivre's Theorem to simplify each of the following, giving your answer in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$:

a) $\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)^{7}$

b) $\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)^{5}$

c) $\left[2\left(\cos\frac{\pi}{5} + \mathrm{i}\sin\frac{\pi}{5}\right)\right]^{11}$

d) $\left[3\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right)\right]^{8}$

e) $\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)^{-7}$

f) $\left[2\left(\cos\frac{4\pi}{9} + \mathrm{i}\sin\frac{4\pi}{9}\right)\right]^{-4}$

Question 2. Use De Moivre's Theorem to simplify each of the following, giving your answer in the form $a + b\mathrm{i}$ where a and b are real constants:

a) $\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)^{15}$

b) $\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right)^{10}$

c) $\left[4\left(\cos\frac{3\pi}{8} + \mathrm{i}\sin\frac{3\pi}{8}\right)\right]^{6}$

d) $\left[\mathrm{i}\left(\cos\frac{5\pi}{4} + \mathrm{i}\sin\frac{5\pi}{4}\right)\right]^{5}$

e) $\left[-3\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)\right]^{6}$

f) $\left[2\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right)\right]^{-4}$

Prerequisite knowledge for Q3: Exponential Form

Question 3. Use De Moivre's Theorem to simplify each of the following, giving your answer in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) $\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)^7$

b) $\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{10}$

c) $\left[3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)\right]^5$

d) $\left[3\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right)\right]^8$

e) $\left[2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^{-4}$

f) $\left[-3\left(\cos\frac{5\pi}{8} + i\sin\frac{5\pi}{8}\right)\right]^{-3}$

Question 4. Use De Moivre's Theorem to simplify each of the following leaving your answer in the form $r(\cos\theta + i\sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$:

a) $(1 + \sqrt{3}i)^7$

b) $(1 - i)^{10}$

c) $(2i)^{20}$

d) $(-8 + 8\sqrt{3}i)^5$

Question 5. Find the value of

$$\frac{1}{(-1+\sqrt{3}\mathrm{i})^4}$$

giving your answer in the form $a+b\mathrm{i}$ where a and b are non-zero real values.

Question 6. Find the value of

$$\frac{1}{(2-2\mathrm{i})^5}$$

giving your answer in the form $a+b\mathrm{i}$ where a and b are non-zero real values.

Question 7.

a) Write $\cos\theta - \mathrm{i}\sin\theta$ in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$

b) Hence simplify $(\cos\theta - \mathrm{i}\sin\theta)^8$

Question 8. Write

$$\left(\cos\frac{\pi}{5} - \mathrm{i}\sin\frac{\pi}{5}\right)^9$$

in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$.

Question 9. Without using an expansion, simplify

a) $(1+\mathrm{i})^8 + (1-\mathrm{i})^8$

b) $(\sqrt{3}+\mathrm{i})^7 + (\sqrt{3}-\mathrm{i})^7$

Question 10. Use De Moivre's theorem to simplify

$$(\cos 9\theta + \mathrm{i}\sin 9\theta)(\cos 6\theta + \mathrm{i}\sin 6\theta)$$

Question 11. Use De Moivre's Theorem to write each of the following in the form $r(\cos\theta+\mathrm{i}\sin\theta)$ where $-\pi<\theta\leq\pi$

a) $\left(\cos\frac{4\pi}{3}+\mathrm{i}\sin\frac{4\pi}{3}\right)\left(\cos\frac{7\pi}{3}+\mathrm{i}\sin\frac{7\pi}{3}\right)$

b) $\left(\cos\frac{17\pi}{4}+\mathrm{i}\sin\frac{17\pi}{4}\right)\left(\cos\frac{5\pi}{4}+\mathrm{i}\sin\frac{5\pi}{4}\right)$

c) $\left(\cos\frac{7\pi}{2}+\mathrm{i}\sin\frac{7\pi}{2}\right)\left(\cos\frac{13\pi}{2}-\mathrm{i}\sin\frac{13\pi}{2}\right)$

Question 12.

$$z=\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)^4\left(\cos\frac{\pi}{3}-\mathrm{i}\sin\frac{\pi}{3}\right)^5$$

a) Express z in the form $r(\cos\theta+\mathrm{i}\sin\theta)$ where $-\pi<\theta\leq\pi$

b) Hence find the value of z

Question 13. Use De Moivre's theorem to simplify

a) $\dfrac{(\cos 9\theta + \mathrm{i}\sin 9\theta)^4}{(\cos 4\theta + \mathrm{i}\sin 4\theta)^6}$

b) $\dfrac{(\cos 5\theta + \mathrm{i}\sin 5\theta)^8}{[2(\cos 3\theta + \mathrm{i}\sin 3\theta)]^6}$

Question 14. Simplify each of the following, giving your answer in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$:

a) $\dfrac{\cos\frac{4\pi}{11} + \mathrm{i}\sin\frac{4\pi}{11}}{\cos\frac{5\pi}{11} - \mathrm{i}\sin\frac{5\pi}{11}}$

b) $\dfrac{\left(\cos\frac{9\pi}{14} + \mathrm{i}\sin\frac{9\pi}{14}\right)^6}{\left(\cos\frac{3\pi}{14} - \mathrm{i}\sin\frac{3\pi}{14}\right)^4}$

Question 15. Use De Moivre's theorem to show that for real values a and b, $(a + b\mathrm{i})^n + (a - b\mathrm{i})^n$ is real for all integers n

Question 16. Find the smallest positive integer value of n, such that $\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)^n$ is a real number.

Question 17.

$$z = \left[3\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right)\right]^n$$

a) Find the smallest positive integer value of n, such that z is a real number.
b) State the value of z for the smallest positive value of n, such that z is a real number.

Question 18.

$$z = \frac{\sqrt{3} - \mathrm{i}}{\sqrt{3} + \mathrm{i}}$$

a) Express z in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$
b) Find the smallest positive integer value of n such that z^n is a real number.

Question 19.

a) Express $2 + 2\mathrm{i}$ in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$

b) Find the smallest positive integer value of n such that $(2 + 2\mathrm{i})^n$ is a purely imaginary number.

Question 20. Given, for integers m and n,

$$z = \frac{\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)^m}{\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)^n}$$

a) Find the smallest positive integer value of n for which z is purely imaginary, and state the corresponding value of m

b) Find the smallest positive integer value of n for which z is real, and state the corresponding value of z

Question 21. The cubic equation $x^3 - px^2 + qx - 4 = 0$ where p and q are positive integers, has roots $\alpha = 1 + \mathrm{i}$, β and γ

a) Find the values of p and q

b) Show that if n is an integer, then

$$\alpha^n + \beta^n + \gamma^n = 2^{\frac{n}{2}+1} \cos\left(\frac{n\pi}{4}\right) + 2^n$$

Question 22. The cubic equation $x^3 - px^2 + 12x - q = 0$ where p and q are positive integers, has roots $\alpha = 1 + \sqrt{3}\mathrm{i}$, β and γ

a) Find the values of p and q

b) Show that if n is an integer, then

$$\alpha^n + \beta^n + \gamma^n = 2^{n+1} \cos\left(\frac{n\pi}{3}\right) + 2^{2n}$$

Prerequisite Knowledge for Q23: Hyperbolic Functions

Question 23.

a) For integer n, show that

$$(\cosh z + \sinh z)^n = \cosh nz + \sinh nz$$

b) Hence determine the exact value of $\cosh(3\ln 2) + \sinh(3\ln 2)$

Question 24. Given that $z = \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}$

a) Write z^2 and z^3 in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$

b) Sketch each of

$$1,\ 1+z,\ 1+z+z^2$$
$$\text{and } 1+z+z^2+z^3$$

on an Argand diagram.

c) State the exact argument of

i) $1+z$

ii) $1+z+z^2$